Java Control Structures

Control structures allow the flow of execution in a program to be directed depending on conditions and repetitions. Java has three main types of control structures:

- **Decision-Making Structures**
- **Looping Structures**
- **Jump Statements (e.g., `break`, `continue`)**

1. Decision Structures

a) `if` Statement

```
int score = 80;
if (score > 70) {
    System.out.println("Passed!");
}
```

b) `if...else` Statement

```
int score = 60;
if (score >= 70) {
    System.out.println("Passed!");
} else {
    System.out.println("Failed.");
}
```

c) `if...else if...else` Chain

```
int score = 85;
if (score >= 90) {
    System.out.println("Grade A");
} else if (score >= 80) {
    System.out.println("Grade B");
} else {
    System.out.println("Grade C or below");
}
```

d) `switch` Statement

```
int day = 3;
switch(day) {
    case 1:
        System.out.println("Monday");
        break;
    case 2:
        System.out.println("Tuesday");
        break;
    default:
        System.out.println("Another day");
}
```

2. Looping Structures

a) `for` Loop

Used when the number of iterations is known.

```
for (int i = 1; i <= 5; i++) {
    System.out.println("Count: " + i);
}
```

b) `while` Loop

Used when the number of iterations is unknown but condition is checked before the loop starts.

```
int i = 1;
while (i <= 5) {
    System.out.println("Count: " + i);
    i++;
}
```

c) `do...while` Loop

Executes at least once. Condition is checked after the loop.

```
int i = 1;
do {
    System.out.println("Count: " + i);
    i++;
} while (i <= 5);
```

3. Nested Control Structures

Loops and decision structures can be nested inside one another.

```
for (int i = 1; i <= 3; i++) {
    if (i % 2 == 0) {
        System.out.println(i + " is even");
    } else {
        System.out.println(i + " is odd");
    }
}
```

4. Jump Statements

a) `break`

Terminates a loop or `switch`.

```
for (int i = 1; i <= 5; i++) {
    if (i == 3) break;
    System.out.println(i);
}
```

b) `continue`

Skips the current iteration.

```
for (int i = 1; i <= 5; i++) {
    if (i == 3) continue;
    System.out.println(i);
}
```